CS420 Sample Quiz 2
(SOLUTION)

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Q1. Consider an input image of size $15 \times 15 \times 3$ (H $\times$ W $\times$ D). The image is passed through a convolutional layer with 8 filters of size 5×5, stride 3 and padding 1.
(1) Calculate the output shape of the convolutional layer. Show the calculation process and write the final answer in the form H $\times$ W $\times$ D.

Solution:

$$H_{\text{out}} = W_{\text{out}} = \frac{15 - 5 + 2 \times 1}{3} + 1 = 5,$$
$$D_{\text{out}} = \text{number of filters} = 8.$$

Final output shape: $5 \times 5 \times 8$.

(2) Calculate the total number of trainable parameters in the convolutional layer, including **bias** terms. Show the calculation process.

Solution: Each filter has size

$$5 \times 5 \times 3 + 1 = 76.$$

Total for 8 filters

$$76 \times 8 = 608.$$

Thus, the convolutional layer has 608 trainable parameters.

Q2. Consider the following $4 \times 4 \times 1$ input:

1	2	-1	3
0	1	2	-2
3	-2	-1	0
2	0	1	-2

The input is passed through the following convolutional layer, with stride 2 and no padding, using the 2×2 filter shown below. Assume the **bias** is -1.

0.5	2
1	-0.5

(1) Calculate the output of the convolutional layer. Show the calculation process.

Solution: With stride 2 and no padding, the output shape of the convolution is

$$\begin{aligned} H_{\text{out}} &= W_{\text{out}} = \frac{4-2+0}{2} + 1 = 2, \\ D_{\text{out}} &= \text{number of filters} = 1. \end{aligned}$$

Let Y denote the output,

$$\begin{aligned} Y_{1,1} &= (1 \times 0.5 + 2 \times 2 + 0 \times 1 + 1 \times (-0.5)) - 1 = 3, \\ Y_{1,2} &= ((-1) \times 0.5 + 3 \times 2 + 2 \times 1 + (-2) \times (-0.5)) - 1 = 7.5, \\ Y_{2,1} &= (3 \times 0.5 + (-2) \times 2 + 2 \times 1 + 0 \times (-0.5)) - 1 = -1.5, \\ Y_{2,2} &= ((-1) \times 0.5 + 0 \times 2 + 1 \times 1 + (-2) \times (-0.5)) - 1 = 0.5. \end{aligned}$$

The convolution output is

$$Y = \begin{bmatrix} 3 & 7.5 \\ -1.5 & 0.5 \end{bmatrix}.$$

(2) The output of the convolutional layer is then passed through a 2×2 max-pooling layer. Calculate the final output after max-pooling.

Solution: Apply 2×2 max-pooling to Y:

$$\max\{3, 7.5, -1.5, 0.5\} = 7.5.$$

The final output after max-pooling is

$$\begin{bmatrix} 7.5 \end{bmatrix}.$$

Q3. Suppose a word2vec model provides the vector representations for the following four words:

$$\begin{array}{ll} \text{word1:} & [2, -1, 0] \\ \text{word2:} & [1, -1, -1] \\ \text{word3:} & [2, 0, -1] \\ \text{word4:} & [0, 1, -2] \end{array}$$

Using **cosine similarity**, determine which word is the odd word out among the four words. Provide justification with calculations.

Solution: First compute the average vector (center of all words):

$$\bar{\mathbf{v}} = \frac{1}{4}\Big([2,-1,0] + [1,-1,-1] + [2,0,-1] + [0,1,-2]\Big) = \left[\frac{5}{4}, -\frac{1}{4}, -1\right].$$

Compute cosine similarity ($S_c(u,v) = \frac{u \cdot v}{\|u\|\|v\|}$):

$$S_C(\text{word1}, \bar{\mathbf{v}}) = \frac{[2,-1,0] \cdot \left[\frac{5}{4}, -\frac{1}{4}, -1\right]}{\sqrt{2^2 + (-1)^2 + 0}\sqrt{(5/4)^2 + (-1/4)^2 + (-1)^2}} = \frac{11/4}{\sqrt{5}\sqrt{21/8}} \approx 0.759,$$

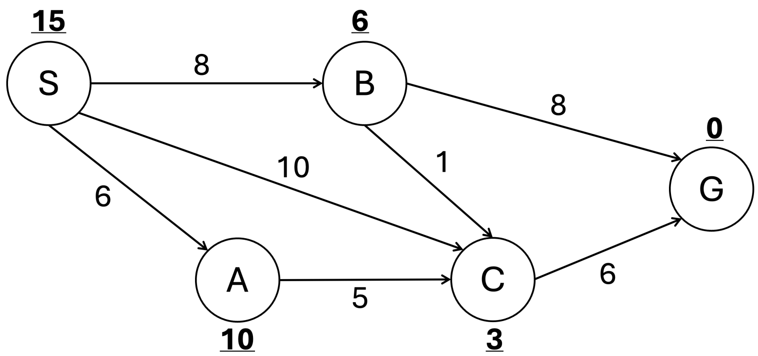
$$S_C(\text{word2}, \bar{\mathbf{v}}) = \frac{[1,-1,-1] \cdot \left[\frac{5}{4}, -\frac{1}{4}, -1\right]}{\sqrt{1^2 + (-1)^2 + (-1)^2}\sqrt{(5/4)^2 + (-1/4)^2 + (-1)^2}} = \frac{5/2}{\sqrt{3}\sqrt{21/8}} \approx 0.891,$$

$$S_C(\text{word3}, \bar{\mathbf{v}}) = \frac{[2,0,-1] \cdot \left[\frac{5}{4}, -\frac{1}{4}, -1\right]}{\sqrt{2^2 + 0 + (-1)^2}\sqrt{(5/4)^2 + (-1/4)^2 + (-1)^2}} = \frac{7/2}{\sqrt{5}\sqrt{21/8}} \approx 0.966,$$

$$S_C(\text{word4}, \bar{\mathbf{v}}) = \frac{[0,1,-2] \cdot \left[\frac{5}{4}, -\frac{1}{4}, -1\right]}{\sqrt{0 + 1^2 + (-2)^2}\sqrt{(5/4)^2 + (-1/4)^2 + (-1)^2}} = \frac{7/4}{\sqrt{5}\sqrt{21/8}} \approx 0.483.$$

Since word4 has the smallest cosine similarity with the average vector, it is the odd word out.

Q4. Use A* Search to solve the following problem from the start state S to goal state G. The **bold and underlined** numbers are the heuristic estimates to the goal, and the numbers on the edges indicate the step costs.



(1) Is the given heuristic admissible? Justify your answer.

Solution:

$$\begin{aligned} h(S) &= 15 \leq \text{min\_cost}(S, G) = 8 + 1 + 6 = 15, \\ h(A) &= 10 \leq \text{min\_cost}(A, G) = 5 + 6 = 11, \\ h(B) &= 6 \leq \text{min\_cost}(B, G) = 1 + 6 = 7, \\ h(C) &= 3 \leq \text{min\_cost}(C, G) = 6, \\ h(G) &= 0 \leq \text{min\_cost}(G, G) = 0. \end{aligned}$$

The heuristic estimate for each state is always less than or equal to the minimum cost from the state to the goal state, so it is admissible. Note that A* with an admissible heuristic is guaranteed to find an optimal solution.

(2) Provide the A* search process step by step.

Solution: A* search:

1. $S(15)$	DQ : $S(15)$	ADD : $A^S(16), B^S(14), C^S(13)$
2. $C^S(13), B^S(14), A^S(16)$	DQ : $C^S(13)$	ADD : $G^C(16)$
3. $B^S(14), A^S(16), G^C(16)$	DQ : $B^S(14)$	ADD : $C^B(12), G^B(16)$
4. $C^B(12), A^S(16), G^C(16)$	DQ : $C^B(12)$	ADD : $G^C(15)$
5. $G^C(15), A^S(16)$	DQ : $G^C(15)$	

Retrieve the path: $G^C \leftarrow C^B \leftarrow B^S \leftarrow S$.
Cost: $6 + 1 + 8 = 15$.

Q5. Suppose a laptop can run in a save mode, a normal mode, or a boost mode. Running in save mode gives lower immediate performance, but it is more likely to keep the laptop cool or help it recover from a hot state. Running in boost mode gives better immediate performance, but it is also more likely to make the laptop overheat. This problem can be modeled as an MDP with two states $S = \{\text{cool}, \text{hot}\}$ and three actions $A = \{\text{save}, \text{normal}, \text{boost}\}$. The transition probabilities $P(s'|s,a)$ and immediate rewards R are given as follows:

s	a	Probability of $s' = \text{cool}$	Probability of $s' = \text{hot}$	Reward R
cool	save	0.85	0.15	6
cool	normal	0.55	0.45	8
cool	boost	0.40	0.60	10
hot	save	0.80	0.20	2
hot	normal	0.30	0.70	3
hot	boost	0.10	0.90	4

Apply Value Iteration to find the optimal policy for this MDP. Assume that the discount factor $\gamma = 1$ and the initial value function $V^0(s) = 0, \forall s \in S$.
(1) Compute the first iteration value function V^1, action-value Q function Q^1, and current policy π^1.

Solution: Value iteration uses

$$Q^{k+1}(s,a) = \sum_{s'} P(s'|s,a)[R(s,a,s') + \gamma V^k(s')], \qquad V^{k+1}(s) = \max_a Q^{k+1}(s,a).$$

First iteration: for $s = \text{cool}$,

$$Q^1(\text{cool}, \text{save}) = 0.85 \times (6 + 1 \times 0) + 0.15 \times (6 + 1 \times 0) = 6,$$

$$Q^1(\text{cool}, \text{normal}) = 0.55 \times (8 + 1 \times 0) + 0.45 \times (8 + 1 \times 0) = 8,$$

$$Q^1(\text{cool}, \text{boost}) = 0.40 \times (10 + 1 \times 0) + 0.60 \times (10 + 1 \times 0) = 10.$$

For $s = \text{hot}$,

$$Q^1(\text{hot}, \text{save}) = 0.80 \times (2 + 1 \times 0) + 0.20 \times (2 + 1 \times 0) = 2,$$

$$Q^1(\text{hot}, \text{normal}) = 0.30 \times (3 + 1 \times 0) + 0.70 \times (3 + 1 \times 0) = 3,$$

$$Q^1(\text{hot}, \text{boost}) = 0.10 \times (4 + 1 \times 0) + 0.90 \times (4 + 1 \times 0) = 4,$$

Therefore,

$$V^1(\text{cool}) = \max(6, 8, 10) = 10, \qquad V^1(\text{hot}) = \max(2, 3, 4) = 4.$$

The current policy π^1 is

$$\pi^1(\text{cool}) = \arg\max_a Q^1(\text{cool}, a) = \text{boost},$$

$$\pi^1(\text{hot}) = \arg\max_a Q^1(\text{hot}, a) = \text{boost}.$$

(2) Compute the second iteration value function V^2, action-value Q function Q^2, and current policy π^2.

Solution: Second iteration: for $s = \text{cool}$,

$$Q^2(\text{cool}, \text{save}) = 0.85 \times (6 + 1 \times 10) + 0.15 \times (6 + 1 \times 4) = 15.1,$$

$$Q^2(\text{cool}, \text{normal}) = 0.55 \times (8 + 1 \times 10) + 0.45 \times (8 + 1 \times 4) = 15.3.$$

$$Q^2(\text{cool}, \text{boost}) = 0.40 \times (10 + 1 \times 10) + 0.60 \times (10 + 1 \times 4) = 16.4.$$

For $s = \text{hot}$,

$$Q^2(\text{hot}, \text{save}) = 0.80 \times (2 + 1 \times 10) + 0.20 \times (2 + 1 \times 4) = 10.8,$$

$$Q^2(\text{hot}, \text{normal}) = 0.30 \times (3 + 1 \times 10) + 0.70 \times (3 + 1 \times 4) = 8.8.$$

$$Q^2(\text{hot}, \text{boost}) = 0.10 \times (4 + 1 \times 10) + 0.90 \times (4 + 1 \times 4) = 8.6.$$

Thus,

$$V^2(\text{cool}) = \max(15.1, 15.3, 16.4) = 16.4, \qquad V^2(\text{hot}) = \max(10.8, 8.8, 8.6) = 10.8.$$

The current policy after the second iteration is

$$\pi^2(\text{cool}) = \arg\max_a Q^2(\text{cool}, a) = \text{boost},$$

$$\pi^2(\text{hot}) = \arg\max_a Q^2(\text{hot}, a) = \text{save}.$$